Breakfast Statistics - Z-Test

communicate:

p = proportion of high school students in Doral Academy High School who eat breakfast daily

$H_0 = p = 0.274$ (27.4%)

$H_a = p > 0.274$

We will be running a one sample z-test for population proportion

$n = 55 \quad x = 24$

$\hat{p} = \frac{24}{55} = 0.436$ (43.6%) $\quad \alpha = 0.05$

conditions:

- we ran a random sample of 55 students, as stated in our survey design
- $55 < 0.10$ (all high school students in Doral Academy)
- $np = 0.274(55) = 15.07 > 10$ ✓

  $n(1-p) = 0.726(55) = 39.93 > 10$ ✓

Since all conditions are satisfied, we can proceed with the z-test

calculate:

$P_0 = 0.274$

$\hat{p} = 0.436$

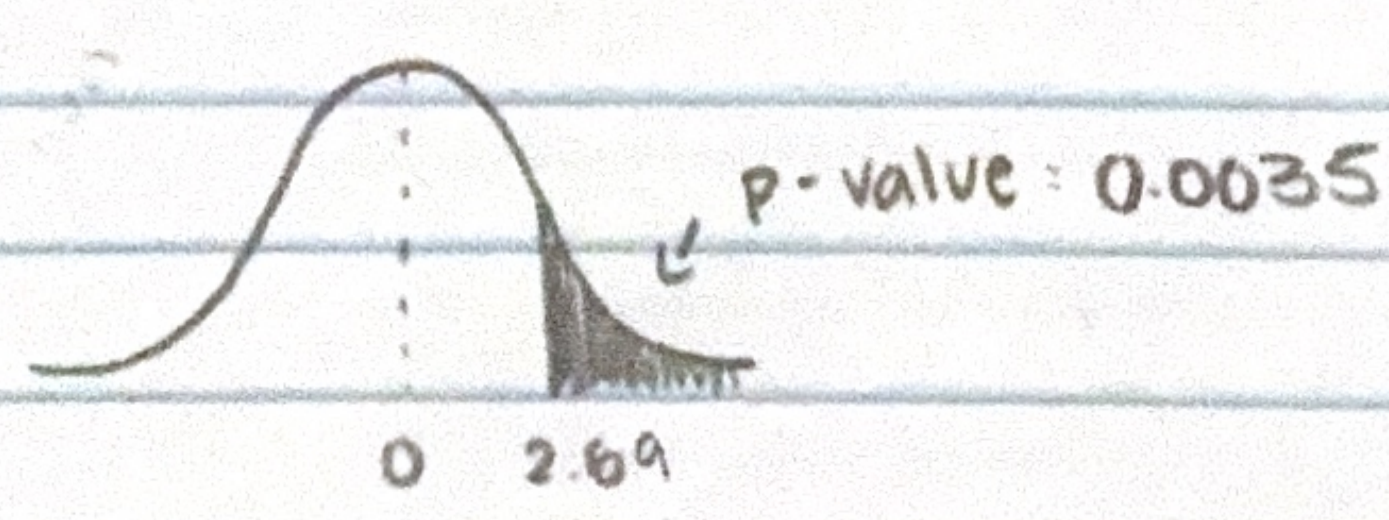
$n = 55$

$\alpha = 0.05$

test statistic $= \dfrac{\hat{p} - P_0}{\sqrt{\dfrac{P_0(1-P_0)}{n}}}$

$$\frac{0.436 - 0.274}{\sqrt{\frac{0.274(0.726)}{55}}} = 2.6937$$

$P(\hat{p} > 0.274, z > 2.6937) = 0.0035$



conclude:

Since our p-value (0.0035) is lower than our significance level (0.05), we can reject the null hypothesis. Therefore, there is sufficient evidence that shows the proportion of high school students at Doral Academy that eat breakfast daily is greater than the national average in the US (0.274)

Breakfast Statistics - Confidence Interval

communicate:

p: proportion of high school students in Doral Academy High School who eat breakfast daily

We will be running a one sample z interval for population proportion

n: 55 x: 24 Ci: 0.95

$\hat{p}$ = 0.436 p = 0.274

conditions:

- we ran a random sample of 55 students, as stated in our survey design
- 55 < 0.10 (all high school students in Doral Academy)
- np: 0.274 (55) = 15.07 > 10 ✓
  n(1-p) = 0.726 (55) = 39.93 > 10 ✓

Since all conditions are satisfied, we can proceed with the confidence interval

calculate:

$\hat{p}$ = 0.436

p = 0.274

n = 55

z^* = 1.96

confidence interval = $\hat{p} \pm z^* \left(\sqrt{\frac{p(1-p)}{n}}\right)$

$0.436 \pm 1.96 \left(\sqrt{\frac{0.274(0.726)}{55}}\right)$

(0.305, 0.567)

conclude:

We are 95% confident that the true proportion of high school students in Doral Academy that eat breakfast daily is between 0.305 and 0.567

Chi-Square Test

chi-square test for independence

o - observed e - expected

	Daily (5)		Not Daily (0-4)		total
9th	6	4.8	5	6.2	11
10th	4	6.55	11	8.45	15
11th	6	6.11	8	7.89	14
12th	8	6.55	7	8.45	15
total	24		31		55

Is there an association between grade level and whether students eat breakfast every day?

H_0: grade level and daily breakfast habits are independent

H_a: grade level and daily breakfast habits are associated

conditions:

- A random sample of students was selected from multiple classrooms at Doral Academy by randomly drawing classroom numbers and student names
- 55 < 0.10 (all high school students at Doral Academy)
  - └ all conditions are satisfied, so we can proceed

calculate:

$$\chi^2 = \sum \frac{(o-e)^2}{e}$$

$$\frac{(6-4.8)^2}{4.8} + \frac{(5-6.2)^2}{6.2} + \frac{(4-6.55)^2}{6.55} + \frac{(11-8.45)^2}{8.45}$$
$$+ \frac{(6-6.11)^2}{6.11} + \frac{(8-7.89)^2}{7.89} + \frac{(8-6.55)^2}{6.55} + \frac{(7-8.45)^2}{8.45}$$

$df = (4-1)(2-1) = 3$

$= 2.8679$

$P(\chi^2 > 2.879) = 0.411$

conclude:

Since our p-value was 0.41 is greater than the significance level of 0.05, we fail to reject the null hypothesis. Therefore, there is not sufficient evidence to conclude that grade level and whether students eat breakfast daily are associated among students at Doral Academy High School